BISL504 DATA VISUALIZATION LAB · WEEK 5

RELYNDIA — Week 5 Summary

Week 5 of the RELYNDIA project establishes a structured, reproducible pipeline for preparing verified datasets before visualization. The work spans three connected stages: **data cleaning and reshaping**, **automated data profiling and quality checking**, and **preparation of visualization-ready outputs**. Together, these stages transform raw verified CSV files into clean, documented, and analysis-ready datasets suitable for interactive Plotly visualizations.

The first script loads verified CSV datasets using Pandas and creates standardized clean DataFrames, preserving the original verified files while producing separate processed datasets in the `clean_data/` directory. Key transformations include standardizing column names such as Financial Year, Revenue, EBITDA, and PAT; extracting numerical years from labels like FY2024; and calculating derived financial metrics including segment revenue share, consumer-facing share, EBITDA margin, and PAT margin. The script also calculates telecom operator market shares from subscriber counts and total wireless subscribers, and computes Reliance Retail productivity metrics such as revenue per store, revenue per square foot, average store size, and retail EBITDA margin.

The financial and telecom datasets are additionally reshaped into long or tidy format for analysis. In wide format, each segment or operator has its own column, which is convenient for human-readable tables. In long format, each observation is represented by a row with fields such as year, segment/operator, and value — a structure particularly convenient for Plotly visualizations.

1

Data Cleaning & Reshaping

Standardize columns, calculate derived metrics, reshape to wide/long format

2

Automated Profiling & Quality Check

Profile four verified datasets, check dimensions, missing values, types, and statistics

3

Visualization-Ready Outputs

Generate seven Plotly charts across three analytical business questions

Scripts 2 & 3 — Profiling, Visualization Engine & Design

Script 2 — Verified Data Profiling & Quality Check

The second script automatically profiles four verified datasets: RIL Financial & Segment Growth, Indian Telecom Market Subscribers, Reliance Retail Footprint, and Indian Economic Indicators. Quality checks performed include dataset dimensions, missing values, duplicate rows, Pandas data types, time horizon, non-null counts, unique values, and numerical descriptive statistics including count, mean, minimum, median, maximum, and standard deviation. The results are compiled into `docs/data_profile.md`, creating a reproducible record of the structure and statistical characteristics of the verified datasets.

Script 3 — Core Plotly Visualization Engine

The third script consumes the cleaned datasets and generates seven core Plotly visualizations organized around three analytical questions. The charts are organized as follows:

- **Reliance diversification:** Line chart, grouped bar, and 100% stacked area — showing revenue trajectories, annual segment comparison, and business-mix transition
- **Telecom market dynamics:** Multi-line and 100% stacked area — showing subscriber trends and operator market-share evolution
- **Reliance Retail expansion:** Dual-axis revenue vs. stores and dual-axis customer count vs. floor area — comparing financial growth with physical footprint and consumer reach

Visualization Design Choices

Line Charts

Used to show trends and trajectories over financial years.

Grouped Bar Chart

Used for direct year-wise comparison between business segments.

100% Stacked Area Charts

Used to show changes in relative composition or market share over time.

Dual-Axis Charts

Used when two related variables have different units and scales, such as revenue and store count or customers and floor area.

A reusable `save_plot()` function saves every visualization as an interactive HTML file and attempts to export a PNG version for static use. Plotly interaction supports hover information, zooming, panning, and legend-based exploration.

End-to-End Pipeline & Week 5 Progress

The three scripts work together as one reproducible workflow rather than as independent pieces of code. The first prepares the data, the second documents its quality and structure, and the third turns the prepared data into visual analytical outputs. The pipeline follows an ETL-style approach: extract verified CSV data, transform it through cleaning and feature engineering, validate and document its quality, reshape it for analysis, and then use Plotly to create interactive visualizations.

01

Verified Datasets

Four verified CSV datasets identified and used as source inputs

02

Quality Profiling

Dimensions, missing values, duplicates, types, and descriptive statistics checked

03

Cleaning & Feature Engineering

Standardized DataFrames, revenue shares, market shares, margins, and Retail productivity metrics calculated

04

Wide → Long Reshaping

Tidy datasets prepared for Plotly analysis

05

Plotly Visualization

Seven core visualizations generated across three business questions

06

HTML + PNG Outputs

Charts exported as interactive HTML files; PNG export attempted for report and presentation use

Week 5 Progress Tracker

Task / Workstream	Progress	Deliverable / Outcome
Verified dataset preparation	✅ Completed	Four verified datasets identified and used as source inputs
Data cleaning & standardization	✅ Completed	Standardized clean DataFrames and processed CSV files in `clean_data/`
Feature engineering	✅ Completed	Revenue shares, market shares, margins, and Retail productivity metrics calculated
Wide → long reshaping	✅ Completed	Tidy datasets prepared for Plotly analysis
Automated data profiling	✅ Completed	Dimensions, missing values, duplicates, types, and descriptive statistics checked
Quality documentation	✅ Completed	`docs/data_profile.md` generated
Plotly visualization engine	✅ Completed	Seven core visualizations generated across three business questions
Interactive HTML outputs	✅ Completed	Charts exported as interactive HTML files
Static PNG outputs	✅ Completed / Optional	PNG export attempted for report and presentation use
Week 5 integration	✅ Completed	Cleaning → validation → visualization pipeline established

Week 5 Takeaway: The project has moved from raw/verified data toward a reproducible, visualization-ready analytical pipeline. Data preparation, data quality documentation, visualization generation, and output organization across `clean_data/`, `docs/`, and `outputs/` directories are all complete. The next practical step is to use the generated visualizations for interpretation and presentation: identify important trends, compare business segments and operators, and connect each chart back to its corresponding analytical question.